

# Surrogate-Aloud: A Human Surrogate Method for Remote Usability Evaluation and Ideation in Virtual Reality

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## ABSTRACT

Virtual and augmented reality offer comparative performance in terms of remote usability testing to lab-based co-located settings. However, direct contact with a researcher is still required to provide setup, troubleshoot, and training. In this paper, we present Surrogate-Aloud as a remote ideation and usability method that establishes a surrogate relationship between participants and a facilitating researcher through video conferencing. The researcher wears a VR headset and shares their viewpoint through video conferencing to a remote participant, who applies think-aloud protocol to express movement and interaction commands to be executed by the researcher, alongside their thought process as they interact with virtual prototypes or scenarios. We conducted a preliminary study to evaluate the Surrogate-Aloud method for remote usability evaluation and ideation of a new instructional technique with volumetric recordings. Results show that Surrogate-Aloud leverages the surrogate's technical expertise and enables sufficient capability to conduct truly remote usability evaluation and ideation.

## CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *Virtual reality*.

## KEYWORDS

Virtual Reality Surrogate, Remote Usability Testing, Think-Aloud

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## 1 INTRODUCTION

Usability evaluation typically involves a facilitator co-located with a participant. Remote usability evaluation does not require the facilitator to be in the same room as the participants [26], thereby allowing participation from people across greater geographical diversity. Previous studies [2, 18, 25, 26], have confirmed comparable effectiveness in uncovering and identifying usability issues between remote usability testing using video conferencing tools and traditional lab-based settings. The use of immersive technology such as virtual and augmented reality (VR/AR) in remote usability testing [4] has also been shown to offer similar performance in terms of the number of usability defects detected by participants. An ideal remote usability evaluation and ideation scenario occurs between researchers and participants when all parties use VR headsets to remotely operate in a shared virtual environment [6] from disperse geographical locations. However, this setup still requires some minimal physical contact with the researcher to assist in configuring and donning headsets on participants, setting up the workstation, and training participants to use the VR system.

In this paper, we present a human surrogate method to conduct remote usability for early stage evaluation and ideation in VR. Surrogates are robotic or virtual representations [14] that can carry out tasks through commands issued by a human user [27]. Rather than direct interaction with VR technology, the participant connects via video conferencing tool to facilitating researchers who wear VR headsets and act as a surrogate for participants in the virtual environment. We apply a verbalization approach used in Think-Aloud Protocol (TAP) as the strategy for participants to issue commands to their surrogate as the surrogate performs interactions in virtual reality on the participant's behalf. Our method is most suitable for lightweight context, sufficient for early feedback and ideation. Our method offers a highly accessible and geographically remote approach for usability and ideation in virtual reality running over consumer grade video conferencing software without requiring additional technology. For researchers, supporting the method only requires the addition of a visual component to allow mouse click interactions and directions into the VR environment.

We conducted a preliminary case study evaluation of Surrogate-Aloud for remote ideation and usability evaluation. We recruited one novice and four experts from VR and circus training domains to explore use cases for a new interaction technique in volumetric data in virtual reality. The results indicate the Surrogate-Aloud provides sufficient capability of interactions for the remote user in VR to successfully generate design ideas and discover usability issues especially for feedback of an early prototype. Our empathetic

surrogate utilizes their technical expertise in VR to insulate our participants from all operationalized procedures, thus reducing system crashes and errors, allowing them to focus more on the usability task at hand.

## 2 RELATED WORK

### 2.1 Think-Aloud Protocol

Think-Aloud Protocol (TAP) has its theoretical roots in cognitive psychology, where a person's verbal utterances are used as a validity to their mental processes. Ericsson and Simon [8] originally defined three levels of verbalization captured from participants, with strict requirement for non-interference from the researchers. Boren and Ramey [3] argue that for usability testing, interventions are often required and do not suit the rigid structure of Ericsson and Simon's definition, instead proposing speech communication theory, where a more conversational environment is created between participant and investigator. Cooke [5] performed a systematic evaluation of verbalization in concurrent think-aloud and defined five verbalization categories: reading, procedure, observation, explanation, and other. Elling and colleagues [7] mapped Cooke's categories to their evaluation, while including an additional two: task related, and filler. The categorization of verbalization in concurrent TAP has attracted interest in its effect and applicability for introspection [15, 22]. The Surrogate-Aloud method utilizes procedural verbalization to communicate participants' intent for the surrogate to execute, thus engaging the participant to partake in concurrent think-aloud in a remote setting. For usability scenarios, subjects are commonly tasked with interacting with evolutionary software, which can introduce complexity and frustration and lead to subjects becoming stuck, sidestepping or avoiding key components [3]. Through Surrogate-Aloud, the interaction is delegated to the technology expert (the surrogate), allowing each role to perform their expected duties without unnecessary cognitive complications.

### 2.2 Surrogate Control – Human Robot Interaction

Human surrogates have been categorized as either being an avatar or an agent in human robot interaction (HRI): the former controlling entity being a human and the latter being a computer algorithm. Surrogates can also be classified by their characteristics, defined by their appearance, corporeal presence, and their agency [21]. Prior work in human-surrogate interaction has focused on the transactional interactions between human users and surrogates [1]. Kim and Welch evaluated the effects of social presence when real humans interact with human-surrogates in augmented reality [14], considering the impact of environmental, social and personal contexts of the surrogate experience. Humanoid robots can mimic virtual surrogates which are controlled by a human user, or be inhabited by a human itself, with their live facial features [16], avatar representation [12] or the body movements being mapped for control [11]. The AMITES framework [21] enables human surrogates control in remote environments, allowing for participants to inhabit surrogates, whether they are life-sized screen displays or humanoid robots. Video conferencing has been shown to be a comparable method for controlling surrogate, yet easier to use as compared to fully immersive virtual reality [17].

## 3 SURROGATE-ALOUD METHOD

Surrogate-Aloud connects remote participants into a virtual environment by sharing the viewpoint of a facilitator wearing a VR headset while issuing verbal movement and interaction commands. It affords opportunities for early prototype ideation and usability testing without the need for multiple VR installations nor encumbering participants with learning premature user interface designs. Despite not directly wearing a VR headset and using VR controllers, through a shared first person view and verbal commands, the participant is in a suitable position to provide usability feedback on menu design, inputs, and output.

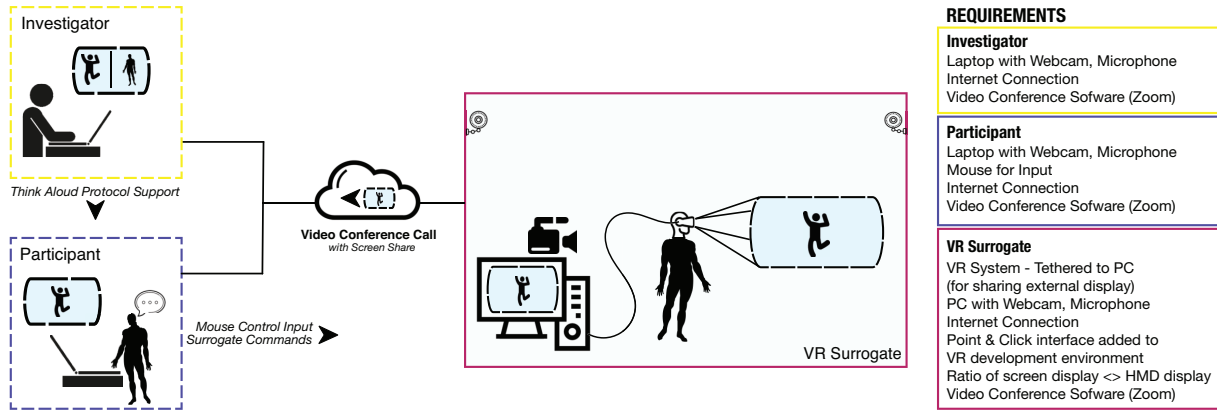
While Surrogate-Aloud is not designed to be used for temporal quantitative measurements from the participant such as in Fitt's law, timed interactions, or physical input, the associated visual design of the virtual environment is a suitable platform for usability and ideation scenarios. In the following sections, we describe the different components of our proposed Surrogate-Aloud method, consisting of the roles of the investigators and the surrogate and the required setup procedure to conduct the Surrogate-Aloud method.

### 3.1 Configuration

The Surrogate-Aloud method design involves three users: the remote participant, the study investigator, and the virtual reality surrogate, connected through video conferencing technology, as shown in Figure 1. The VR surrogate is the only person wearing a VR headset in a room-scale facility. The participant and investigator only require a computer terminal to access the video conferencing tool (i.e. Zoom, Teams [24, 30]). A breakdown of the requirements for a Surrogate-Aloud session to be conducted are provided in Figure 1. The surrogate shares their VR viewpoint on their PC in real time through the video conferencing software's built in screen-share functionality. The participant verbally communicates with the investigator and the surrogate. The role of the surrogate is to act as a proxy for all interactions and actions that the participant wishes to complete during the session.

The role of the investigator is to support the ongoing verbalization of the participant's internal thoughts, as required by Think-Aloud protocol. This is achieved through neutral compact probing to further elicit additional information regarding the current visualization being experienced through the virtual reality surrogate, while delivering acknowledgements throughout conversations. The VR surrogate does not converse, acting only as silent listener, physically executing commands delivered by the remote participant.

The participant is tasked with thinking aloud and providing commands to the VR surrogate to perform the required actions inside the virtual space, as if they were embodying the surrogate. The protocol is communicated clearly and succinctly to the participant at the start of the session. The Surrogate-Aloud protocol allows both explicit and casual command sets to be provided to the surrogate. Explicit commands involve a set of instructions to direct the surrogate to perform all actions, such as "walk forward, press the menu button, select play option, change speed to 50%, press the close button". A more casual command set allows requests to be provided with an emphasis on the surrogate interpreting and completing the command; "The silver object in front of you, can you please get it to play at half speed?" A strength of the



**Figure 1: Surrogate-Along Setup with VR surrogate, Investigator and Participant in three separate locations. The remote communication is achieved through video conferencing software with screen-share.**

Surrogate-Along configuration is its ability to be adapted to the needs of the underlying research. Usability testing may require more explicit requests to understand underlying cognitive issues in system design, while ideation scenarios may be better suited to abstracting interactions from the participant’s decision-making process.

Surrogate control is reliant on voice commands through the video conferencing software. To provide agency, a point and click input is provided to the participant to allow more effective use of deictic phrases during command issuing and TAP. Through the video conferencing software, mouse control is provided to the participant who can click on their 2D screen, generating an image plane ray-cast into the virtual environment (Figure 2a). The method for calculating and conveying the correct screen point to world coordinate is achieved through an understanding of the VR technology being used, and the ratio of the environment being displayed on the Surrogate’s screen. The monitor display is a ratio of the VR Surrogate’s total view, commonly displaying only a single eye (e) of a stereoscopic experience. The ray-cast is therefore calculated with a transformed mouse click into the shared view:

$$Tx = ((0.5f * (1.0f - e.w)) * e.w) + (e.w * mouse(x));$$

$$Ty = ((0.5f * (1.0f - e.h)) * e.h) + (e.h * mouse(y));$$

This transformed cursor can then be applied to calculate the correct ray into the environment and update the position and rotation of the virtual object accordingly. This process affords the participant the ability to indicate an explicit point of reference in the environment, as part of a command, or as a conversational aid. Combined with TAP, the point and click action allows communication using deictic terms such as ‘here’ and ‘there’ to occur without associated ambiguity.

As a method to acknowledge success of movement or action, the surrogate can perform non-verbal acknowledgement gestures using virtual hand models, in line with TAP [3, 15]. These gestures can include pointing with both hands, grabbing objects, or a thumbs up (Figure 2b). The visualization forms two important contributions; the surrogate is provided with visual feedback to perform their own successful actions inside the virtual space, while also providing clear feedback to the participant of the intended movement. For

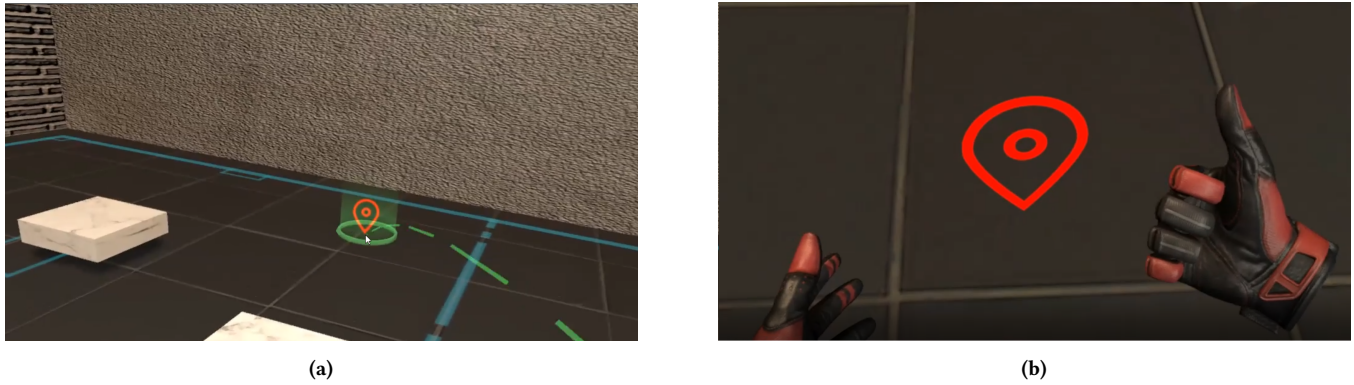
instance, if the surrogate is confused with a command, they can relay a response through shrugging, with both palms open and facing upwards. The control granted over the human surrogate can be adapted to meet the needs of the protocol; either responding only to direct commands or operating under some form of autonomy to meet the requests of the remote participant.

### 3.2 Usage and Benefits

Surrogate-Along offers a platform for researchers to elicit qualitative analysis on early prototypes from remote participants. It offers three unique benefits over the conventional laboratory design: (1) geographical limitations are removed for both researchers and participants, (2) interactions can be abstracted away to reduce the cognitive effort imposed on participants completing the required task, and (3) researchers can perform actions which are not important, allowing participants to focus their attention on areas of necessity for the underlying session.

The Surrogate-Along method can be applied without geographical limitations being imposed on the availability of participants. By applying this method, evaluations can be conducted in virtual environments without the need for in-person interactions, allowing geographically disperse participants to be involved in their own surrounds without being encumbered by VR technology.

Surrogate-Along offers a unique approach for reducing the impact of system errors during usability evaluation or ideation. During conventional in-person testing, a participant will need to be trained on the basic functional operations of a VR system in order to complete the required task set; which can lead to potentially encountering issues specific to VR technology, rather than the usability scenario at hand. As an alternative, the virtual reality surrogate’s actions encapsulate this process into a black box; the participant thinks aloud their intended action, and is presented with the expected output, without the cognitive cost of interacting with the underlying system. The protocol allows the surrogate to assume expert control over system functionality while granting agency for the participant inside the virtual environment. Additionally, early prototypes can be evaluated or ideated without the need to generate a set of interactions suitable for an external party to operate;



**Figure 2: (a) Participants can direct the surrogate to locations of interest using deictic phrases or aid in conversational commands using the point and click interface, transmitting a screen position through video conferencing, and presenting a geospatial pointer for the VR surrogate. (b) The Surrogate can respond through gestures to convey successful completion of requested commands.**

complex actions are abstracted away from potential system breaking scenarios. For usability evaluation, due to the sharing of the first-person perspective, the participant still retains the opportunity to provide valuable feedback throughout the interaction.

#### 4 PRELIMINARY EVALUATION

We conducted a remote ideation scenario for an interaction technique using volumetric recordings in virtual reality to understand the application of our method in practice. Our preliminary evaluation aimed to understand three aspects of Surrogate-Aloud: 1) the effectiveness of Surrogate-Aloud in ideation and usability; 2) the impact of surrogate control on the user experience; and 3) the practice of Think-Aloud technique during surrogate interaction.

We applied Surrogate-Aloud method to ideation and prototype usability sessions with remote users through video-conferencing to explore a novel interaction technique with volumetric recordings. We set up a configuration of the Surrogate-Aloud method as described in Section 3.1 between two remote researchers and a participant. As an ideation domain, we implemented an interaction technique with volumetric recordings as a method for creating instructional technology in VR/AR. We recorded a sample of five volumetric holograms, demonstrating different types of movements and skills. During the session, the participants were asked to investigate the structure of the holograms through the surrogate, ideate on use cases for hologram instructional techniques, and to role play learning the physical movement illustrated by the holograms. The session structure was administered by the remote investigator.

From the sessions, we collected the number of ideas, and scenarios generated by participants and self-reflections on their experiences throughout the session. We were interested in the remote participant's experience through the VR screen-share, the usage of the surrogate-aloud control protocol, and the extent to which the participant experienced the virtual environment; therefore we asked them to complete a series of questionnaires regarding their experiences, including sense of presence [29] and a modified sense of embodiment [23]. The data collection session took approximately one hour. We recruited participants through a student mailing list

at our university and social groups. We selected three participants with varying experiences with VR technologies and further invited two experts in movement training.

We divided our findings into different aspects according to the aims of the evaluation as stated above. The video conference platform for the study was Zoom. All sessions were recorded and automatically transcribed using Zoom cloud server. Errors in the transcription were manually corrected by two authors. Transcription was imported into NVivo for thematic coding.

##### 4.1 Ideation and Method Usability

Using Surrogate-Aloud, our participants successfully identified usage scenarios for volumetric recordings and instructional techniques throughout the movement analysis tasks. We counted 62 scenarios identified in a variety of domains. Participants uttered scenarios throughout the session, either unprompted or as probed by the investigator when they were interacting with the holograms. Through the session, we can observe that our participants could infer use cases beyond the current state of the prototype, which indicates a successful use of VR surrogate in our Surrogate-Aloud for early feedback and ideation.

##### 4.2 Surrogate Control

We are interested in understanding if the participants feel sufficient sense of agency through the surrogate control, with the tools and protocol afforded by Surrogate-Aloud. For movement control where the participants indicated a new location or viewpoint for the surrogate to move to in the virtual room, we counted 134 instances of this request. For all participants, we noticed a shift from a position of command to cooperation towards the surrogate. The verbal movement requests ranged from direct commands, such as 'go over there', 'step to the left', to more reflective intentions, such as 'I'm going to go to the middle one, please', and cooperative requests, such as '(let's) assume a view where you can see both holograms aligned', which requires problem-solving skills for the surrogate to find a location for that view. These requests fit in the procedural verbalization types as outlined by Cooke [5] and Elling et al. [7].

For movement, the surrogate used two methods: physical walking for distance fewer than three steps away, and teleportation. It was up to the surrogate to determine the best travel method based on participants' requests. A recent study of the usability of VR locomotion methods by Mayor et al. [19] showed that teleport is not the most preferred method with low intuitive score. In effect, Surrogate-Aloud through VR surrogate abstracts usability issues with locomotion and interaction. We counted 42 instances of participants asking the surrogate to pick up, move, and place virtual objects. The complexity and usability consideration of virtual controllers have been compared in both virtual reality [9] and surrogate robotic telepresence [28]. Yet, our participants did not require any prior training to operate and interact with virtual objects through the surrogate. Surrogate control extends to other types of full body interaction as well. One participant requested the surrogate to kneel and crouch down to examine the lower part of the volumetric hologram, while another was also interested in a viewpoint looking up to explore how the torso moved. These observations show that despite being limited to only a teleconferencing tool, the participants were engaged with body movements and agency within the virtual environment through VR surrogate.

Across our participants, different tones and approaches were used to describe their reactions and reasoning through Think-Aloud. The participants experienced 'second-hand' interactions with the virtual environments through the surrogate; yet, while thinking aloud, many used first-person pronouns to reflect on the virtual experience. We collected a modified sense of embodiment [13] and presence questionnaire at the end of the session, with the statement 'I felt as if the researcher's body might be my body' receiving a mean score of 67.6 (SD=16.5) and 'The movements I saw as performed by the researcher make it seemed to be my own movements' receiving a mean score of 67 (SD=16.3). Presence statements such as 'I feel present in the virtual space' and 'I was completely captivated by the virtual world' received high mean scores of 80 and 72.6. Overall, we are confident that Surrogate-Aloud via VR surrogate sufficiently convey the awe, agency, and embodiment of VR technology for remote usability testing.

### 4.3 Think-Aloud

The communication protocol with the surrogate expanded as the participant became more familiar with the environment. Landmarks or points of interest in the virtual environment were employed in directional control. Statements referring to virtual points of interest alerted the investigator to the participant's process of reasoning and provided an opportunity to unpack the participant's intentions and thought process through Think-Aloud by asking such questions as 'Why are you interested in that area?' In a sense, verbalization of surrogate control requests provides opportunities to investigate users' intentions through Think-Aloud.

Moreover, we noticed that the surrogate command became compounded with a series of requests to set up a configuration of holograms suitable to explore the scenario at hand. Complex surrogate commands became a form of delegation where the participants expected that their intended actions in the virtual world were 'taken care' of by the surrogate. In a sense, the participants were unburdened from the hardware, controls, interfaces, and could focus their

effort into ideation or usability evaluation. The surrogate absorbed all the operationalized actions during the test session to free up the participants for the usability or ideation task at hand.

## 5 DISCUSSION

Our study has evaluated the remote usability Surrogate-Aloud method for an ideation scenario of volumetric recordings. In this section, we reflect on the intricacies of the method based on our experience of the study.

### 5.1 Empathetic Surrogate

There were many instances where a request was never uttered or implied but was inferred by the surrogate based on the participant's current actions, such as the surrogate autonomously assuming a slightly adjusted viewpoint than the one chosen by the participant. Nonetheless, the participants did not feel like they lost their agency as an inquirer in the virtual space, as shown by the high rating on the presence and embodiment questionnaire. The surrogate acts as a proxy to encourage participants' curiosity in exploring the ideation or usability evaluation scenario.

Our Surrogate-Aloud method employs an empathic human surrogate who can anticipate, interpret, infer the tone and intention of the participants, to facilitate remote usability evaluation and ideation. Exploration of Surrogate-Aloud interaction with other users or agents in the virtual environment could provide interesting insights. Moreover, we have not explored the experience of the surrogate; but we could argue that the surrogate, as an HCI researcher, is afforded a unique viewpoint into the participants' actions and thought processes in a more intimate way than an observing investigator would have.

### 5.2 Verbalization through Surrogate-Aloud – A Rogue Approach

Prior research into think-aloud protocol focused on identifying the different verbalization categories [5, 7, 10], in which procedure is described as a self-report of participants activities and intentions. In the studies by Elling et al. [7] and Cooke [5], procedure made up of 14-20% of all verbalizations. In our study, we did not attempt to classify verbalization, but rather focus on the interactions between the participants and the surrogate. Our reporting of the participants requests for movement, viewpoint, and body interaction categorically fit into the procedure category by Cooke [5] and Elling et al. [7], thus providing a natural application of TAP for remote usability and ideation. Moreover, the classic Think-Aloud protocol [8] prescribes a hands-off approach towards the participant, to avoid contamination of thought data [3]. Other researchers adopt a more lenient approach with Interactive Think Aloud [20] that allows the researcher to request and prompt for information. In contrast, surrogate is an intimate connection, as our participants commented in many instances 'I feel like I was in <surrogate's name>'s shoes'. Our Surrogate-Aloud method goes rogue by introducing the surrogate as a third party and pulls the participant as close to the surrogate as possible. The participant is off-loaded from operational processes to complete ideation tasks without additional effort affecting verbalization. We also did not find any instances of technical glitch or error or failure of the system throughout the study, except for

comments on the quality of the internet connection on the participants' end. It is unsure if we could attribute this to the robustness of our prototype or the technical expertise of our VR surrogate. Instead, our 'empathetic surrogate' abstracts these errors away to allow our participants to focus more on the usability and ideation task.

## 6 CONCLUSION

We present a remote ideation and usability method called Surrogate-Aloud that connects a participant to a human surrogate in virtual reality via video conferencing. The surrogate has direct access to virtual reality hardware and performs interactions and inquiries in a virtual environment at the participant's command, with the assistance of a research investigator. Through Think-Aloud protocol, the participant verbalizes their movement and interaction commands to the surrogate as procedural verbalization, thus benefiting from the operational expertise of the surrogate throughout the session. Surrogate-Aloud provides the benefits of fewer technical errors while encouraging thinking aloud for participants. Our preliminary study shows that Surrogate-Aloud provides sufficient capability of virtual reality interactions, presence, and embodiment, while minimizing the technological exposure for participants. The role of the researcher as an empathetic surrogate can anticipate, interpret, infer the tone and intention of the participants, to facilitate remote usability evaluation and ideation. Future work into the experiences of the surrogate could provide further insight into this new remote usability evaluation and ideation method.

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